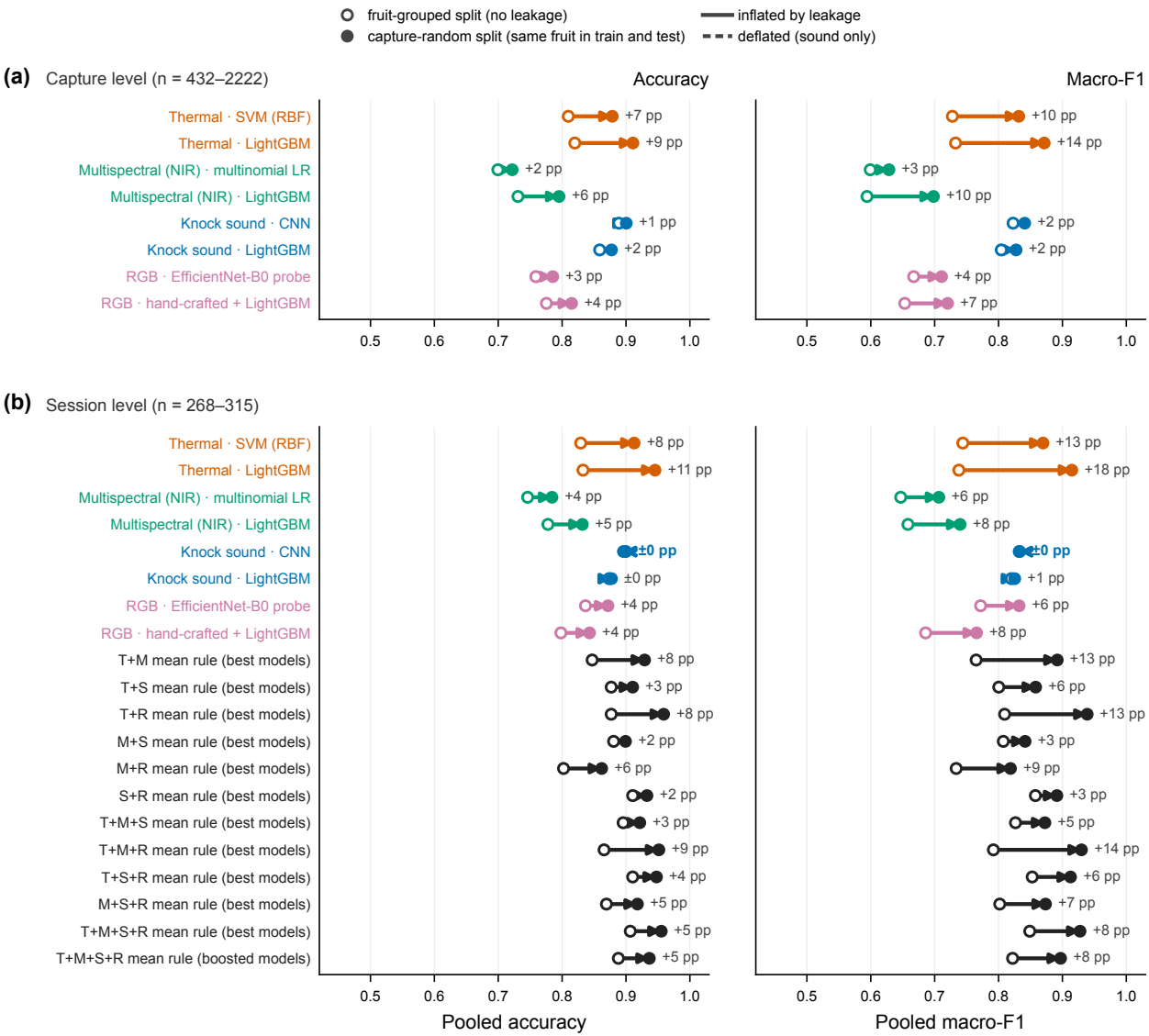

fruit-grouped split (no leakage)
capture-random split (same fruit in train and test)
inflated by leakage
deflated (sound only)
(a)
Capture level (n = 432–2222)
Accuracy
Macro-F1
Thermal · SVM (RBF)
Thermal · LightGBM
Multispectral (NIR) · multinomial LR
Multispectral (NIR) · LightGBM
Knock sound · CNN
Knock sound · LightGBM
RGB · EfficientNet-B0 probe
RGB · hand-crafted + LightGBM
+7 pp
+9 pp
+2 pp
+6 pp
+1 pp
+2 pp
+3 pp
+4 pp
+10 pp
+14 pp
+3 pp
+10 pp
+2 pp
+2 pp
+4 pp
+7 pp
0.5
0.6
0.7
0.8
0.9
1.0
(b)
Session level (n = 268–315)
Thermal · SVM (RBF)
Thermal · LightGBM
Multispectral (NIR) · multinomial LR
Multispectral (NIR) · LightGBM
Knock sound · CNN
Knock sound · LightGBM
RGB · EfficientNet-B0 probe
RGB · hand-crafted + LightGBM
T+M mean rule (best models)
T+S mean rule (best models)
T+R mean rule (best models)
M+S mean rule (best models)
M+R mean rule (best models)
S+R mean rule (best models)
T+M+S mean rule (best models)
T+M+R mean rule (best models)
T+S+R mean rule (best models)
M+S+R mean rule (best models)
T+M+S+R mean rule (best models)
T+M+S+R mean rule (boosted models)
+8 pp
+11 pp
+4 pp
+5 pp
±0 pp
±0 pp
+4 pp
+4 pp
+8 pp
+3 pp
+8 pp
+2 pp
+6 pp
+2 pp
+3 pp
+9 pp
+4 pp
+5 pp
+5 pp
+5 pp
+13 pp
+18 pp
+6 pp
+8 pp
±0 pp
+1 pp
+6 pp
+8 pp
+13 pp
+6 pp
+13 pp
+3 pp
+9 pp
+3 pp
+5 pp
+14 pp
+6 pp
+7 pp
+8 pp
+8 pp
0.5
0.6
0.7
0.8
0.9
1.0
Pooled accuracy
Pooled macro-F1